# Lecture - 2

# Chapter 2

# DIGITAL IMAGE FUNDAMENTALS

# Structure of the Human Eye

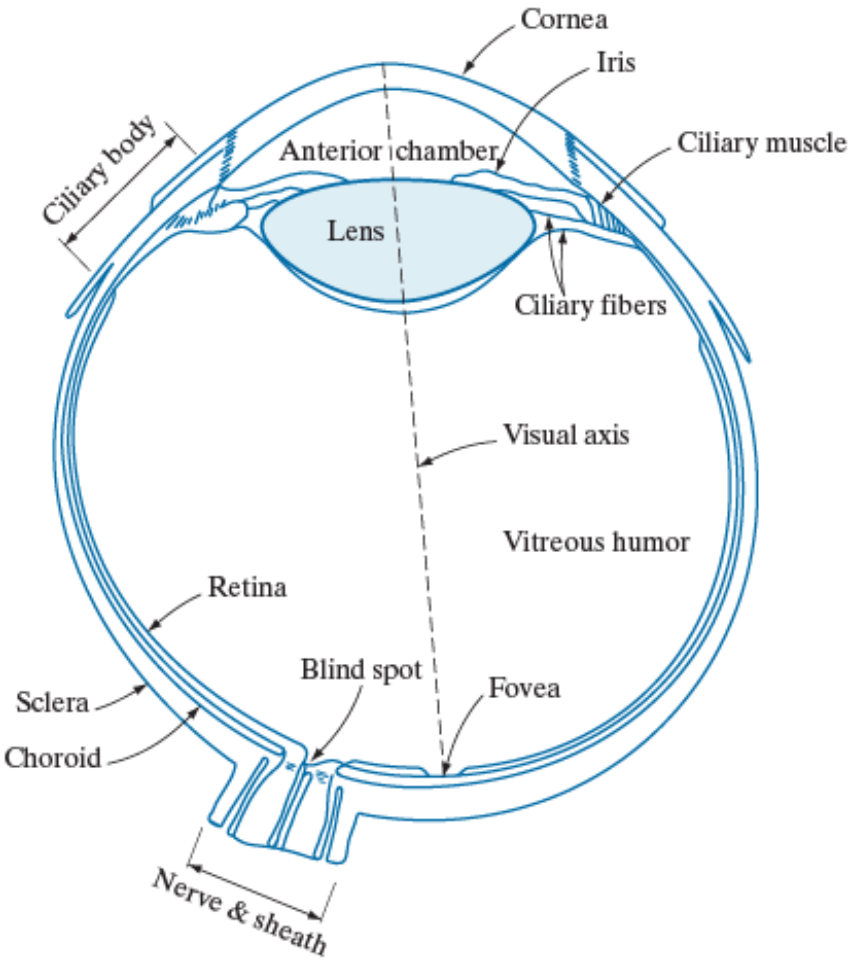


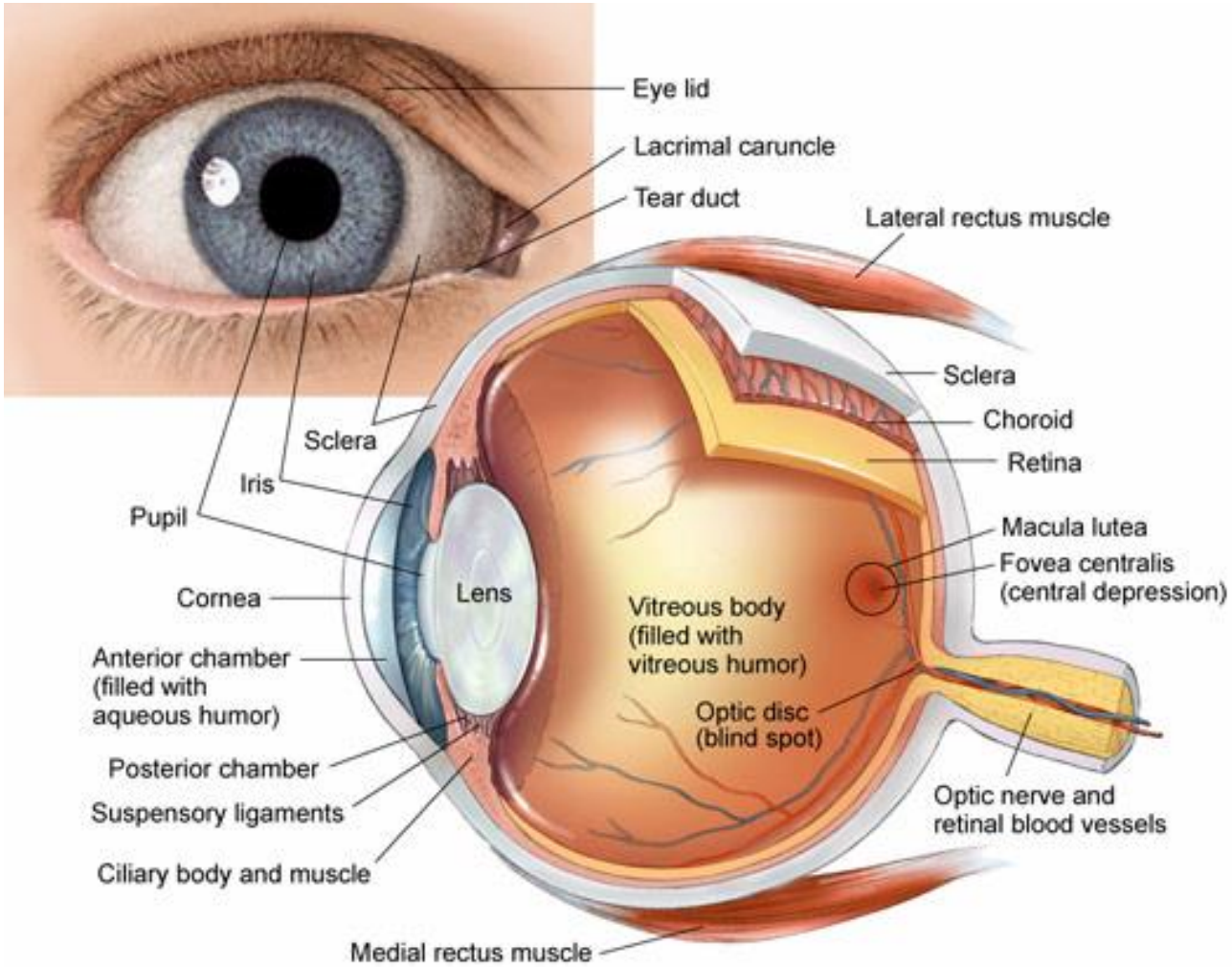

**FIGURE 2.1** Simplified diagram of a cross section of the human eye.

# Structure of the Human Eye

Light enters the eye through the transparent cornea, passing through the pupil at the center of the iris.

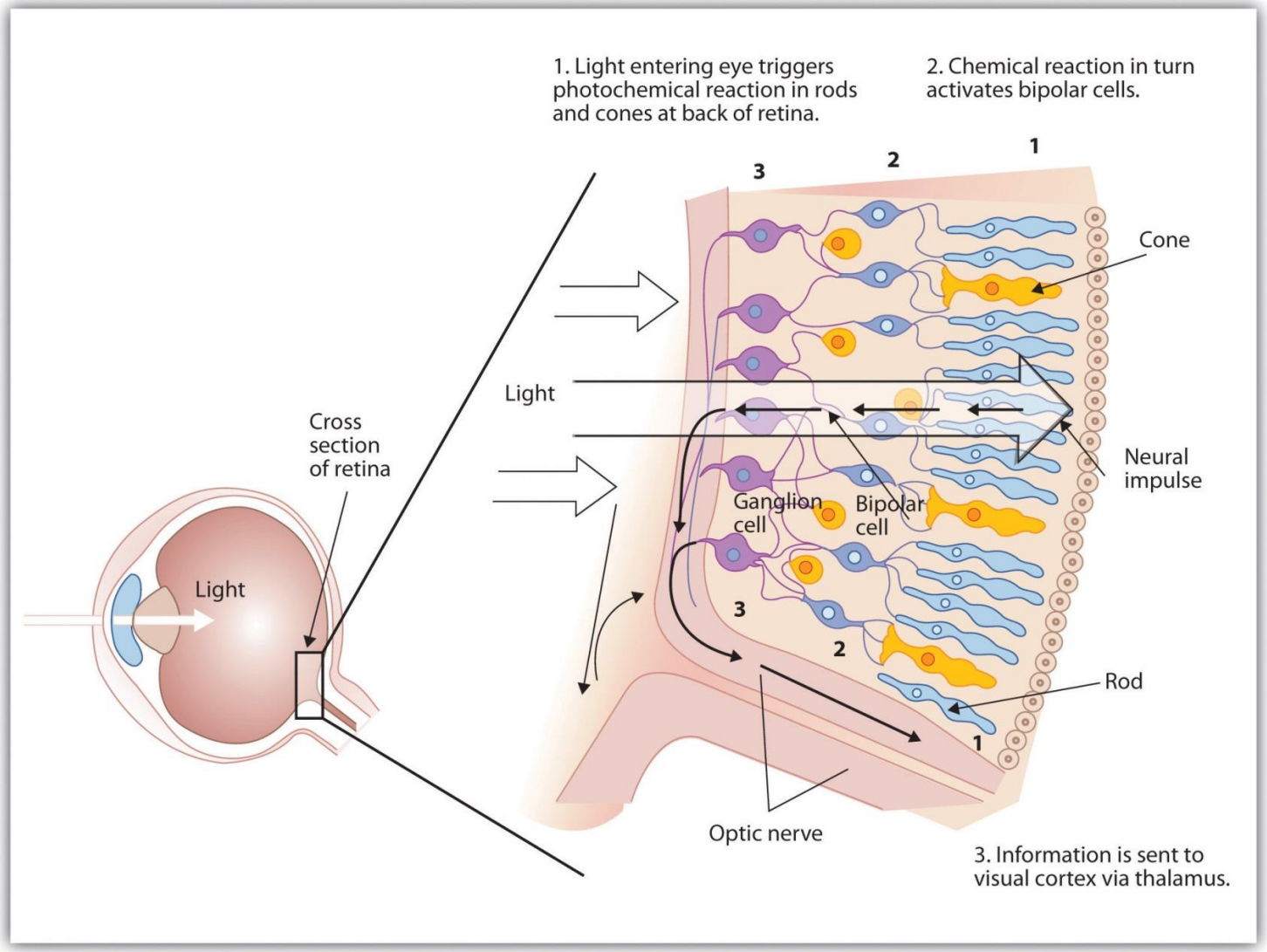
The lens adjusts to focus the light on the retina.

Receptor cells on the retina send information via the optic nerve to the visual cortex.




**Right Eye (viewed from above)**

# Structure of the Human Eye



# Structure of the Human Eye

Retina contains light receptors: **Cones & rods**

| **Cones:** | **Rods** |
| --- | --- |
| 6 to 7 million | 75 to 150 million |
| located mainly in central part of retina (fovea) | Distributed on a wide region of the retina |
| Sensitive to color | No color vision, responsible for low light vision |
| Can resolve fine details | Gives overall picture of field of view |
| Cone vision: photopic or bright-light vision, | Rod vision: scotopic or dim-light |

# Structure of the Human Eye

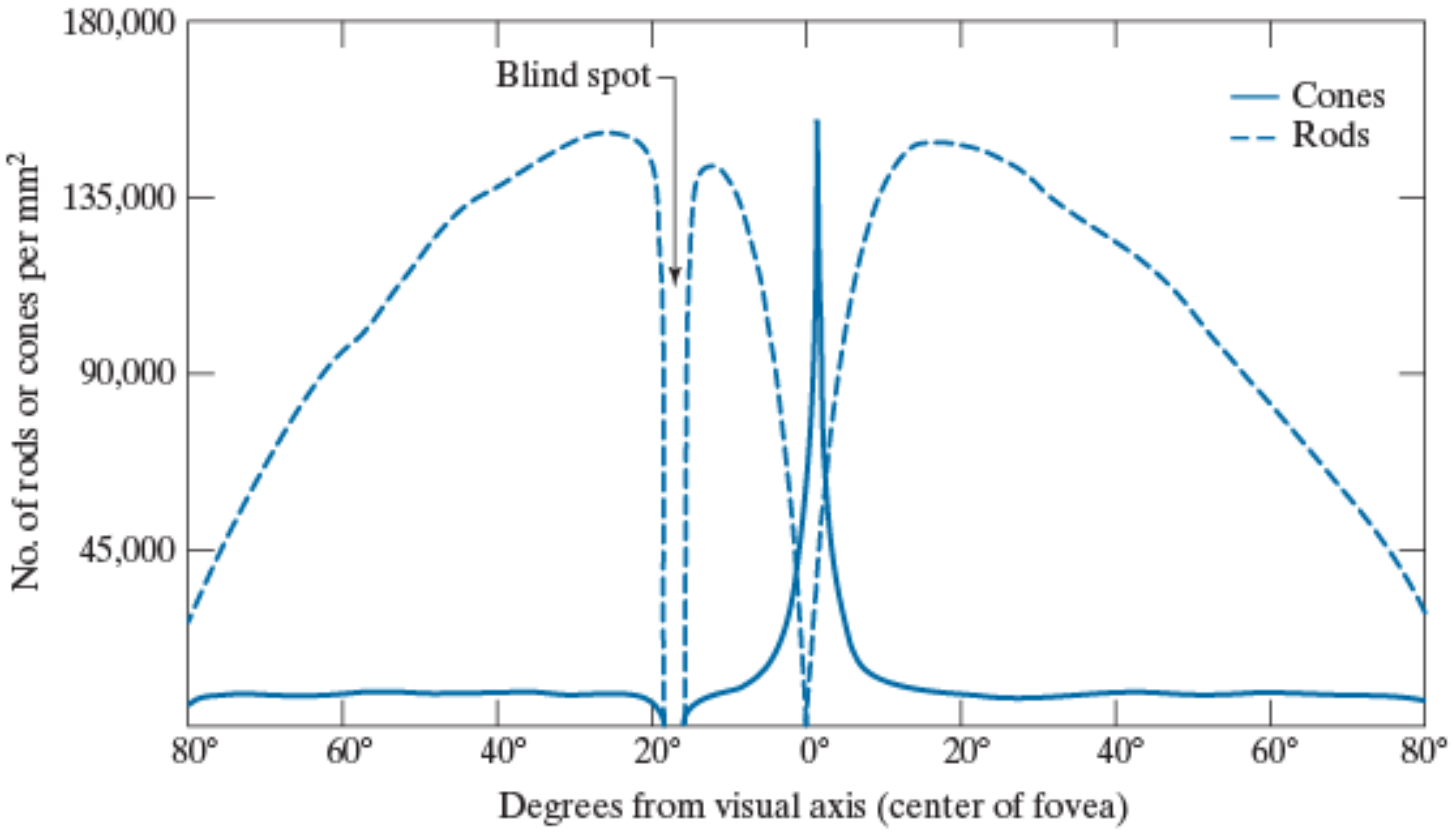



**FIGURE 2.2** Distribution of rods and cones in the retina.

# Structure of the Human Eye

**Blind Spot:** a region of retina without receptors, optic nerves go through this part.

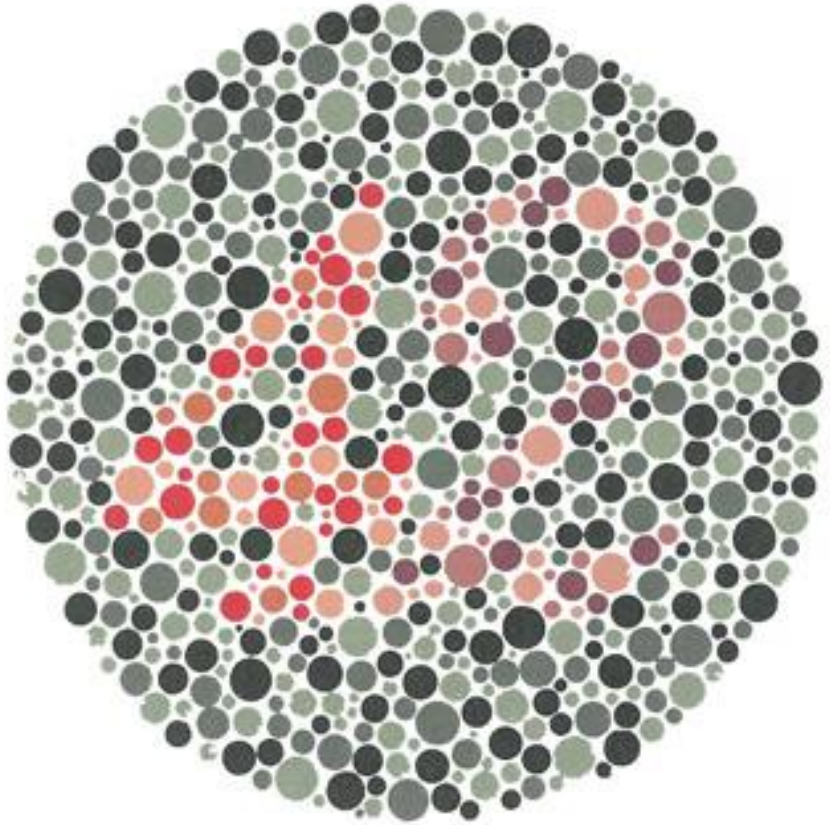
**Fovea:** An area of about 1.77 $mm^2$ (a circular region of diameter 1.5 mm)

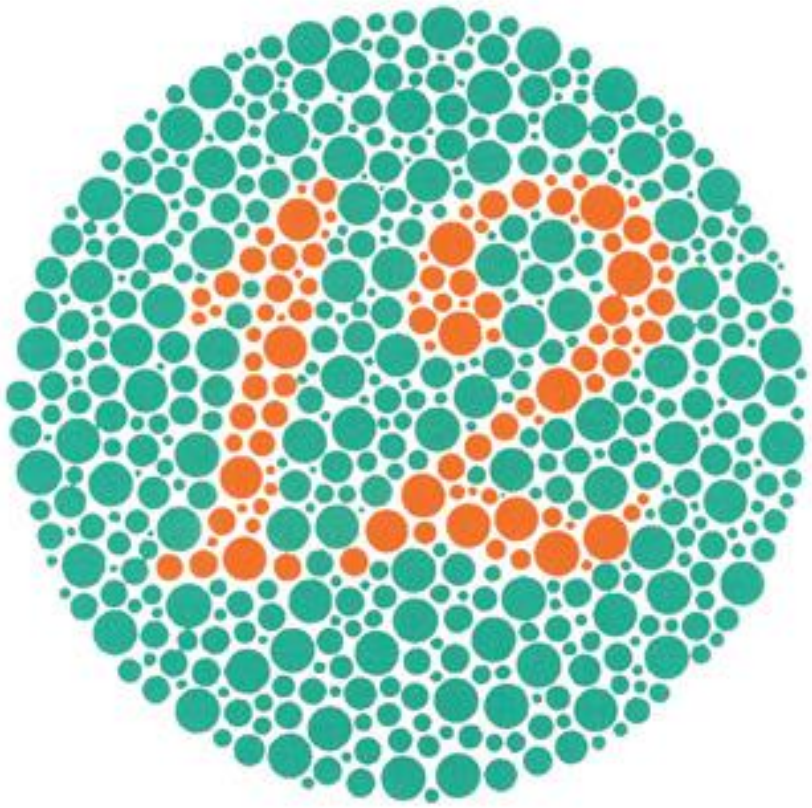
Density of cones on the Fovea = 150,000 elements per $mm^2$

Total number of cones on the Fovea approximately = 265,000

Modern electronics imaging chips exceed this number by a large factor

# Color Test





# Image Formation in the Eye

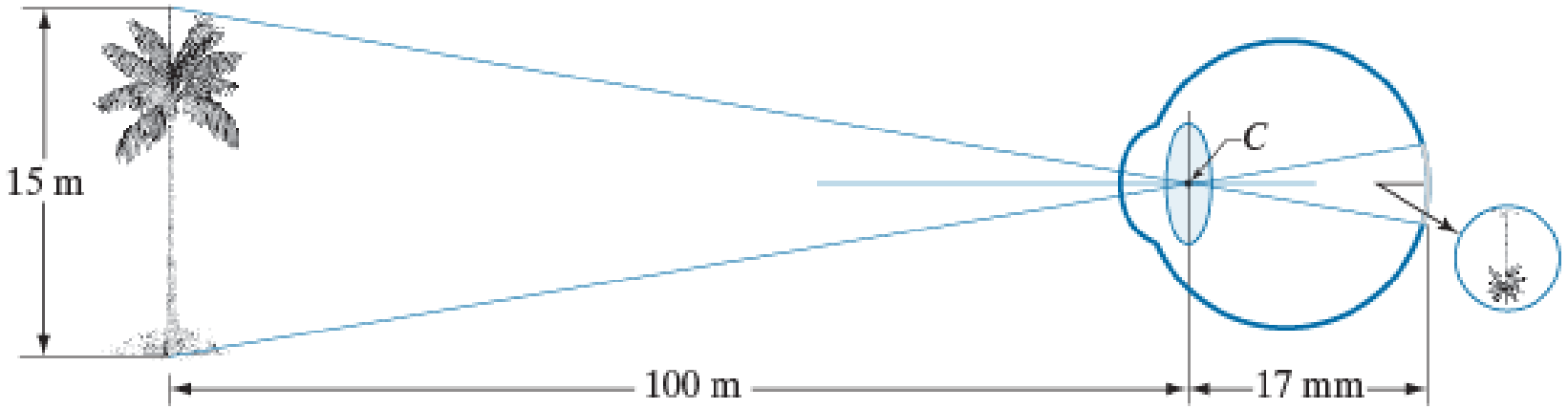



**FIGURE 2.3** Graphical representation of the eye looking at a palm tree. Point *C* is the focal center of the lens.

The image appears upside down on the retina.

$$15/100 = h/17$$

$$h = 2.5 \text{ mm}$$

# Brightness Adaptation & Discrimination

- The dynamic range of light intensity to which eye can adapt is enormous – (on the order of $10^{10}$) from the scotopic threshold to the glare limit
- HVS can not operate over the entire range simultaneously.
- It accomplishes large variations due to brightness adaptation.
- Brightness Ability of the eye to discriminate between changes in brightness at any specific adaptation level.

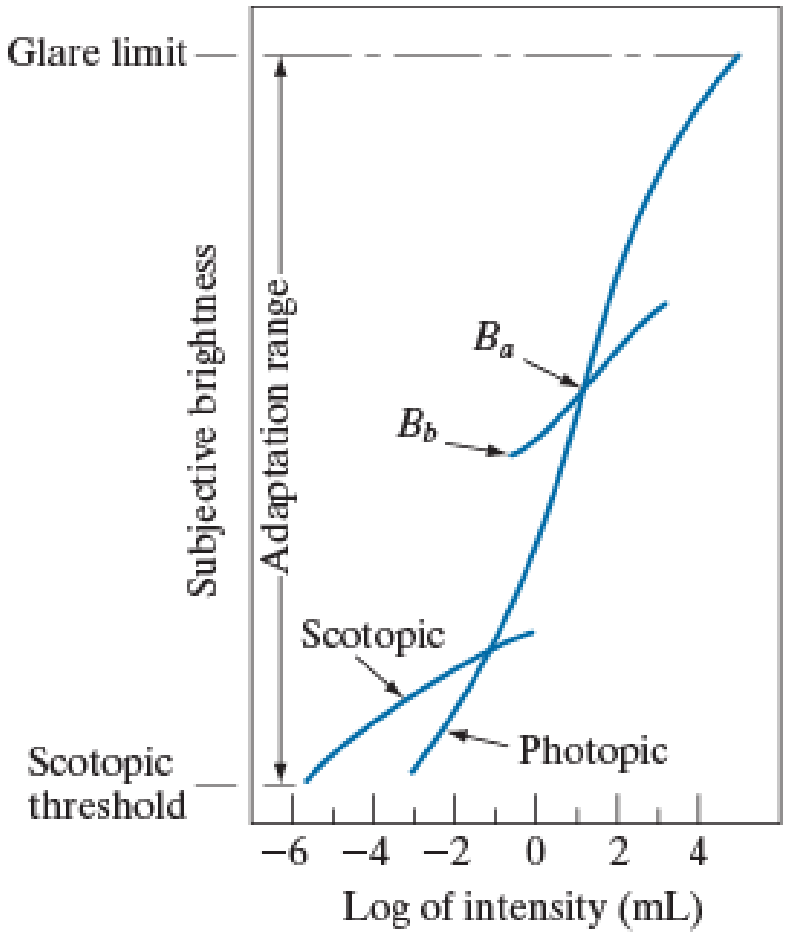



**FIGURE 2.4** Range of subjective brightness sensations showing a particular adaptation level, $B_a$.

# Weber Experiment for Brightness Discrimination

I : Intensity at some point
ΔI : Increment of illumination
If ΔI is not bright, the subject says “no”
If ΔI is strong enough, the subject says “yes”
Weber ratio : ΔI / I
ΔI : Increment discriminable
A small Weber ratio means good brightness

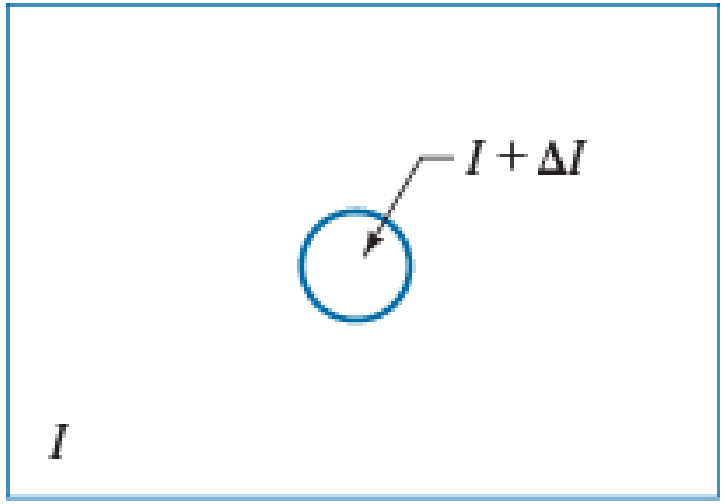



**FIGURE 2.5** Basic experimental setup used to characterize brightness discrimination.

# Weber Experiment for Brightness Discrimination

- Brightness Discrimination is the ability of the eye to discriminate between changes in brightness at any specific adaptation level.

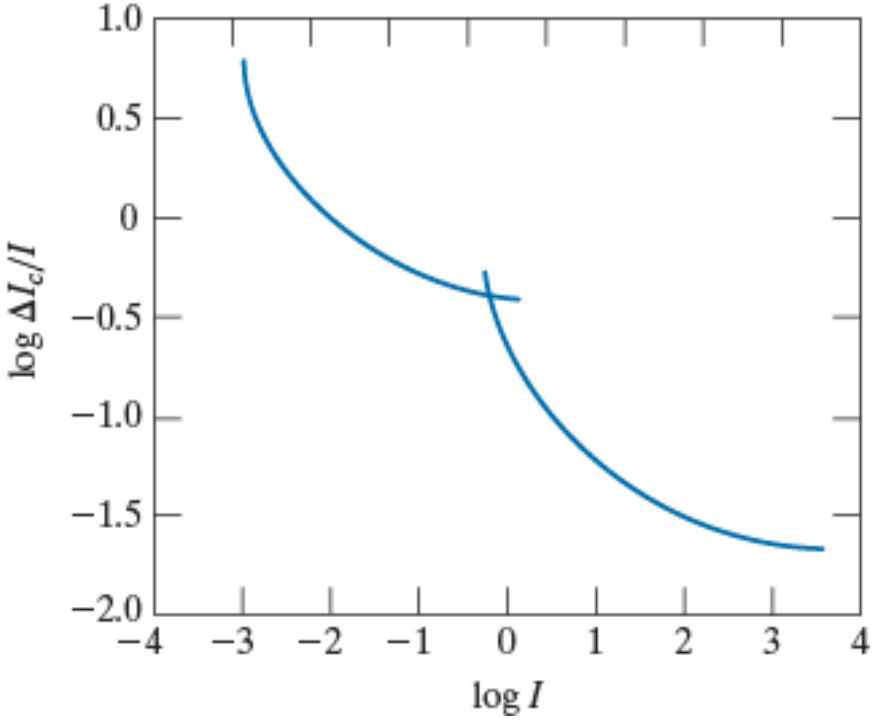
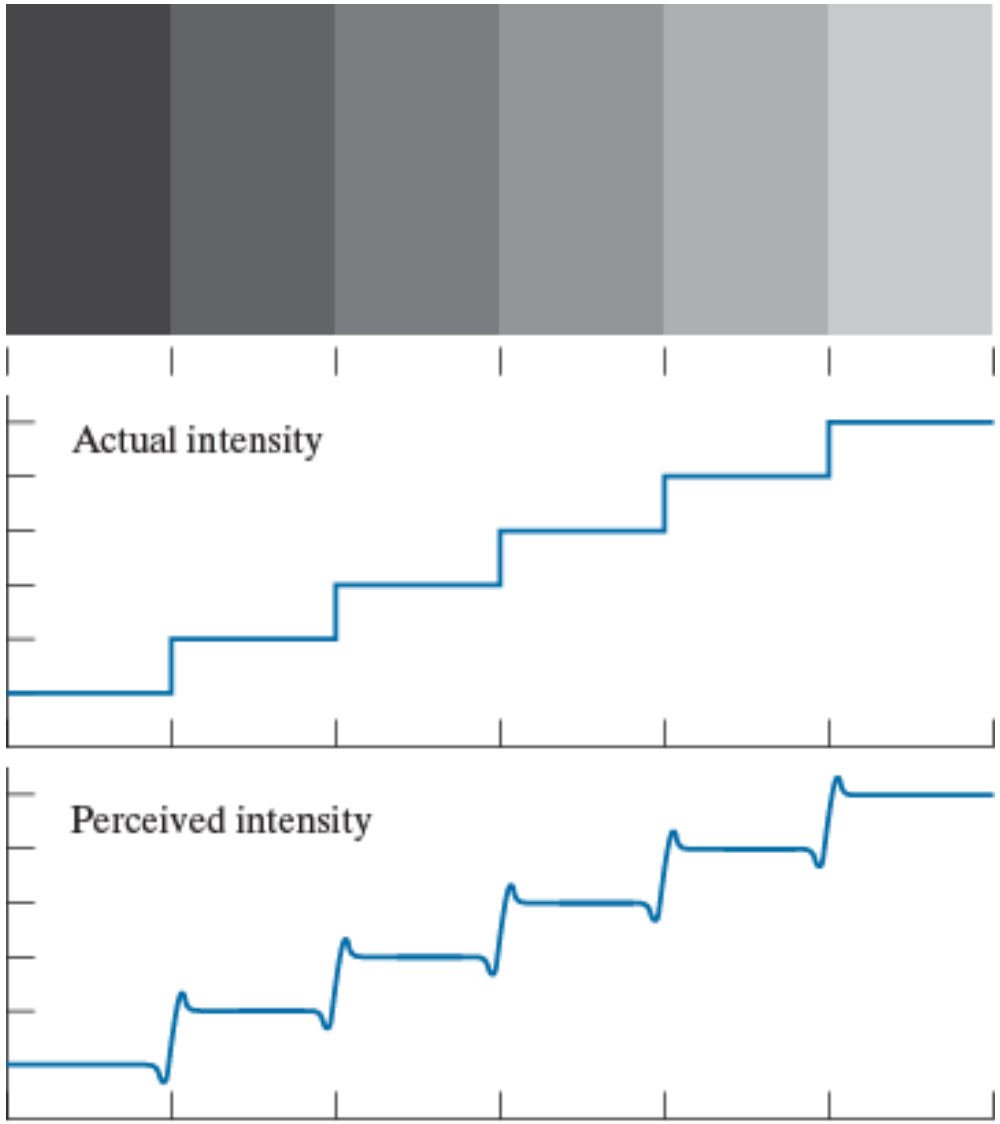



**FIGURE 2.6** A typical plot of the Weber ratio as a function of intensity.

# Mach Band Effect

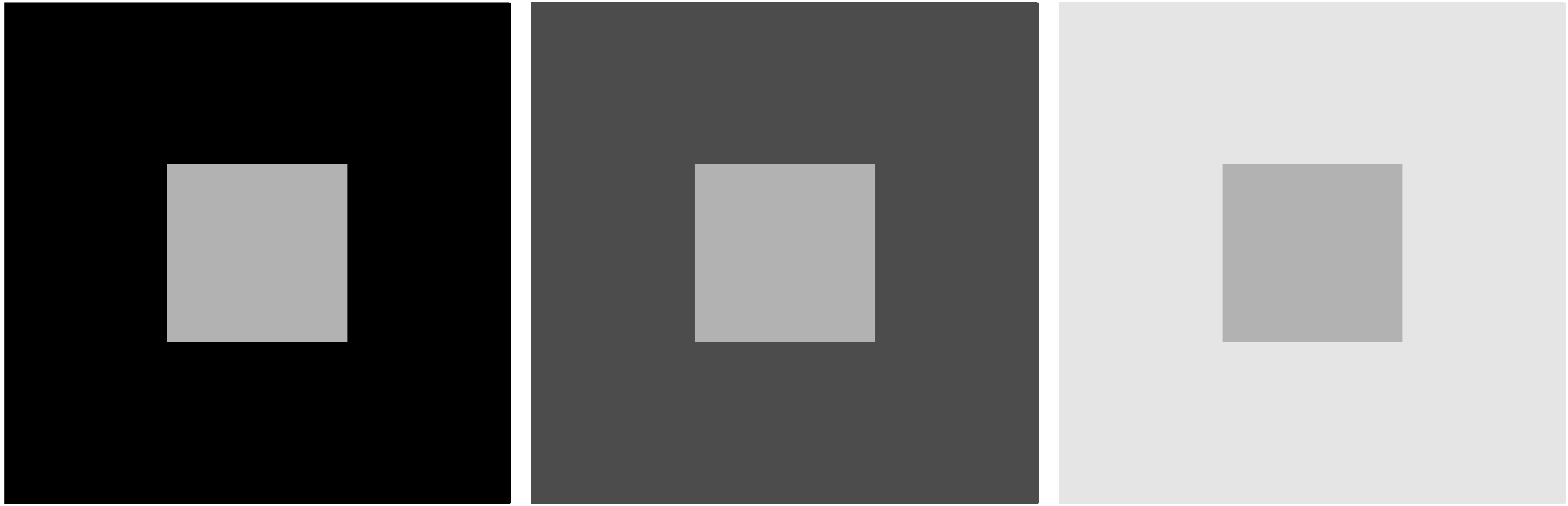
- Perceived brightness is not a simple function of actual intensity.




a
b
c

**FIGURE 2.7** Illustration of the Mach band effect. Perceived intensity is not a simple function of actual intensity.

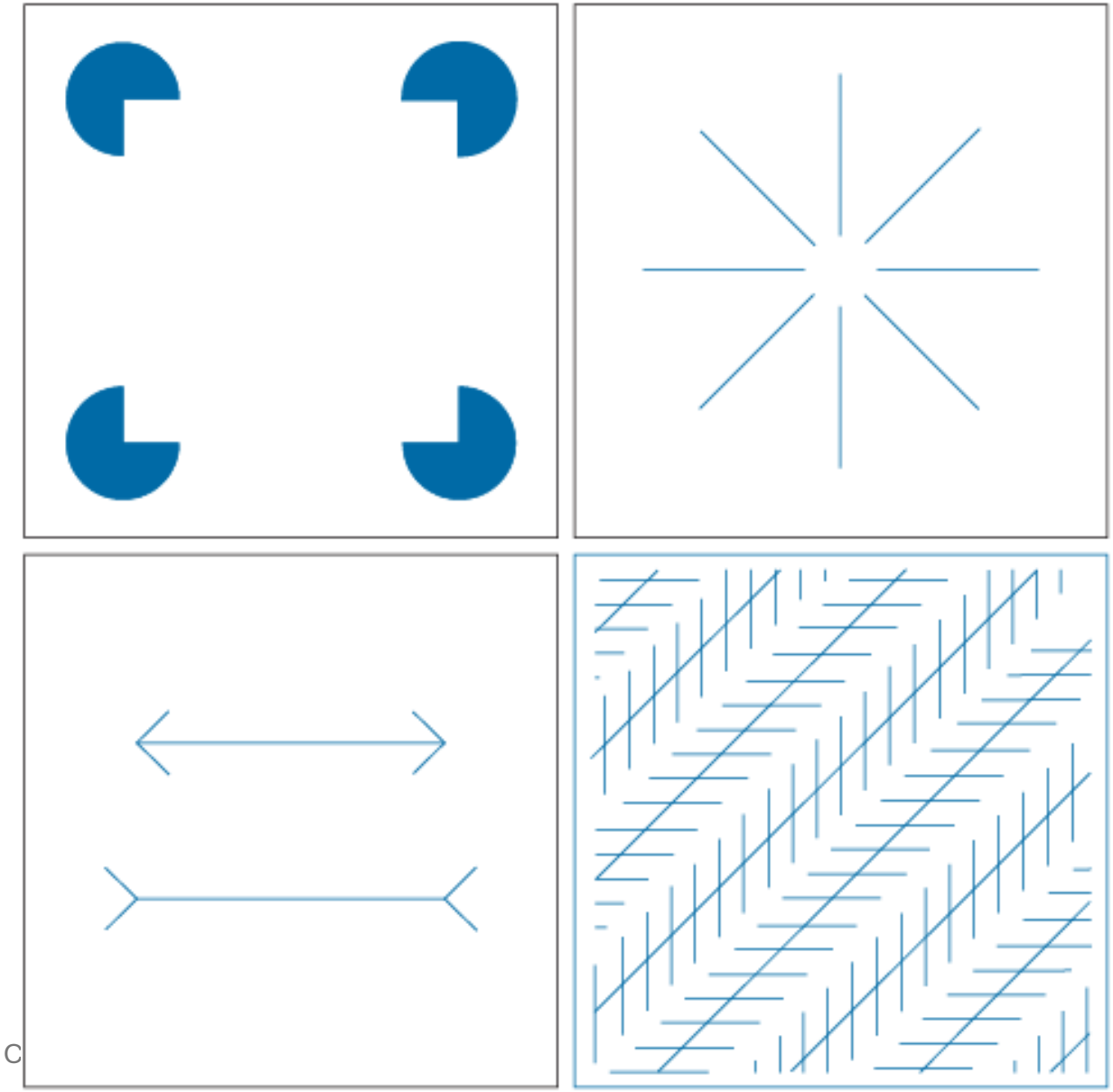
# Simultaneous Contrast



a b c

**FIGURE 2.8** Examples of simultaneous contrast. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

# Optical Illusion



a b
c d

**FIGURE 2.9** Some well-known optical illusions.

# Visible Band of Electromagnetic Spectrum

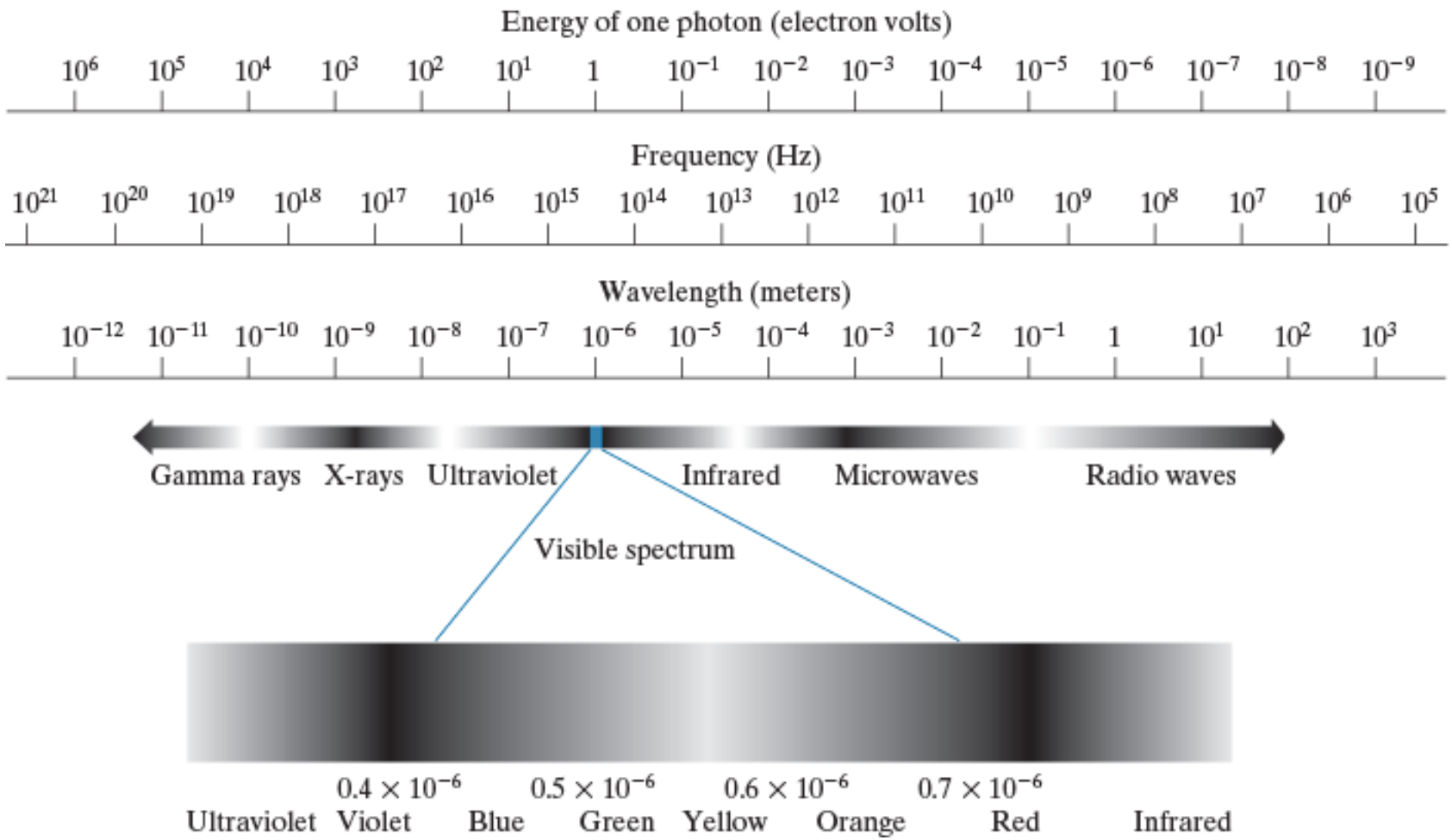
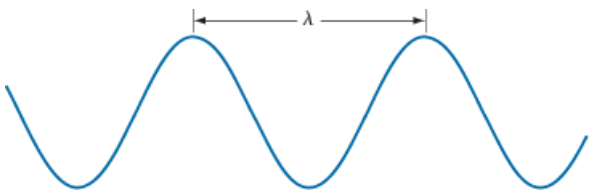



**FIGURE 2.10** The electromagnetic spectrum. The visible spectrum is shown zoomed to facilitate explanations, but note that it encompasses a very narrow range of the total EM spectrum.

**FIGURE 2.11** Graphical representation of one wavelength.



# Radiance, Luminance & Brightness

**Radiance** is the total amount of energy that flows from the light source (Watts).

**Luminance** (measured in lumens lm), gives a measure of the amount of energy an observer perceives from a light source. In case of far infrared, luminance is zero.

**Brightness** is a subjective descriptor of light perception that is practically impossible to measure.

# Sensors

- Sensors are used to transform illumination energy into digital images.

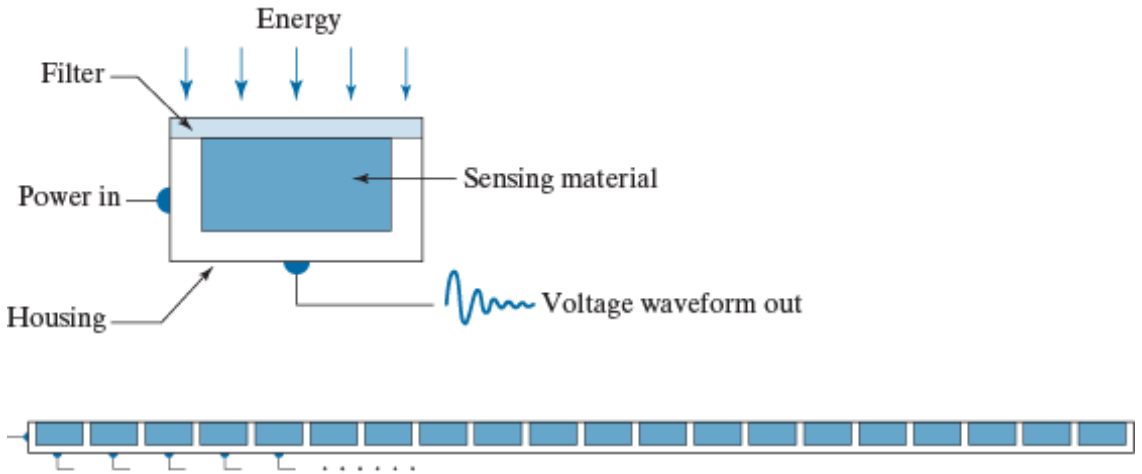
- In principle, if a sensor can be developed that is capable of detecting energy radiated by a band of the EM spectrum, we can image events of interest in that band.

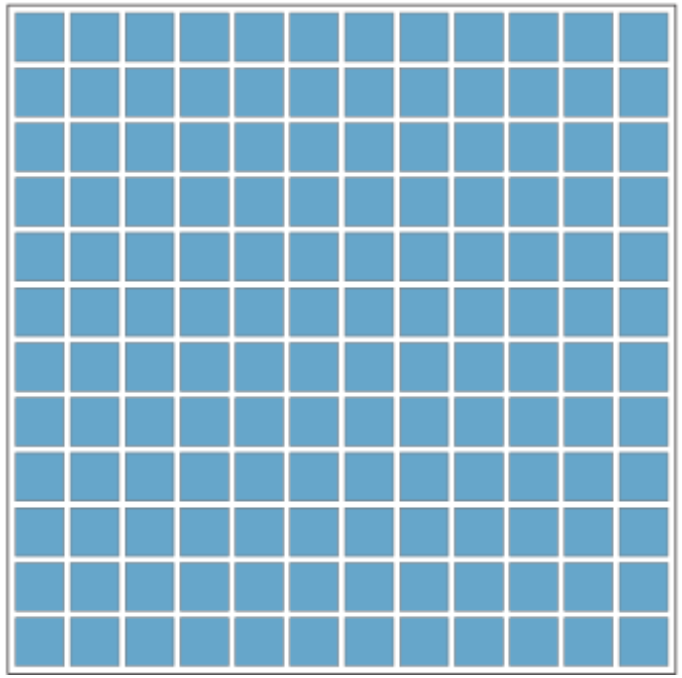
- The wavelength of an EM wave required to “see” an object must be of the same size as or smaller than the object.

  Size of water molecule has a diameter on the order of $10^{-10}$m. To study water molecule, a source emitting in soft x-ray region is required.

- This limitation + physical properties of the sensor material establishes the fundamental limits on the capability of imaging sensors, such as visible, infrared and other sensors in use today.

# Types of Sensors



a
b
c

**FIGURE 2.12**
(a) Single sensing element.
(b) Line sensor.
(c) Array sensor.

# Image Acquisition using Single Sensor

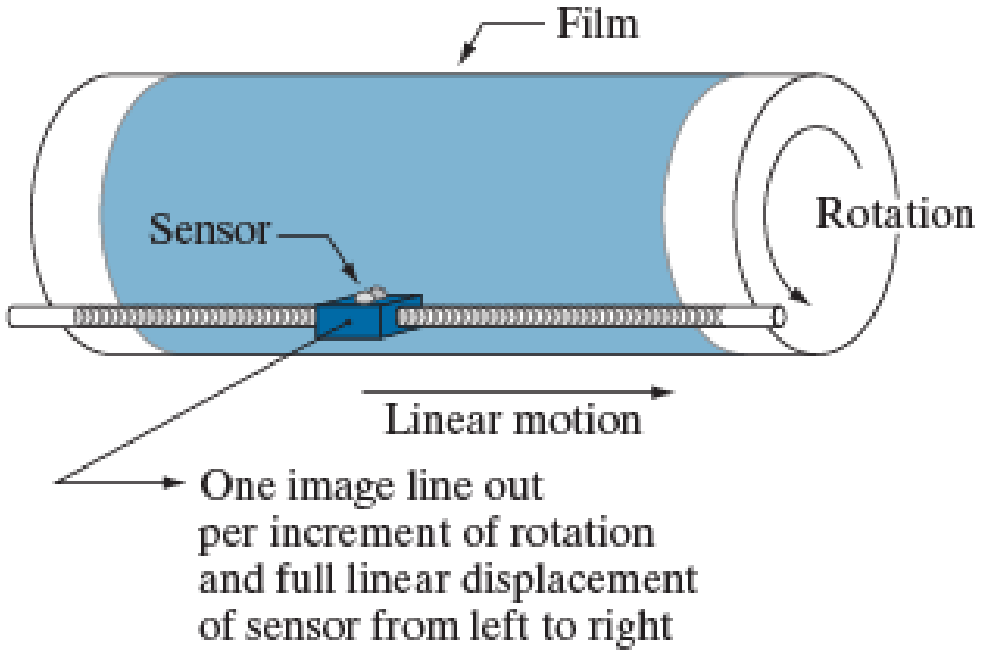
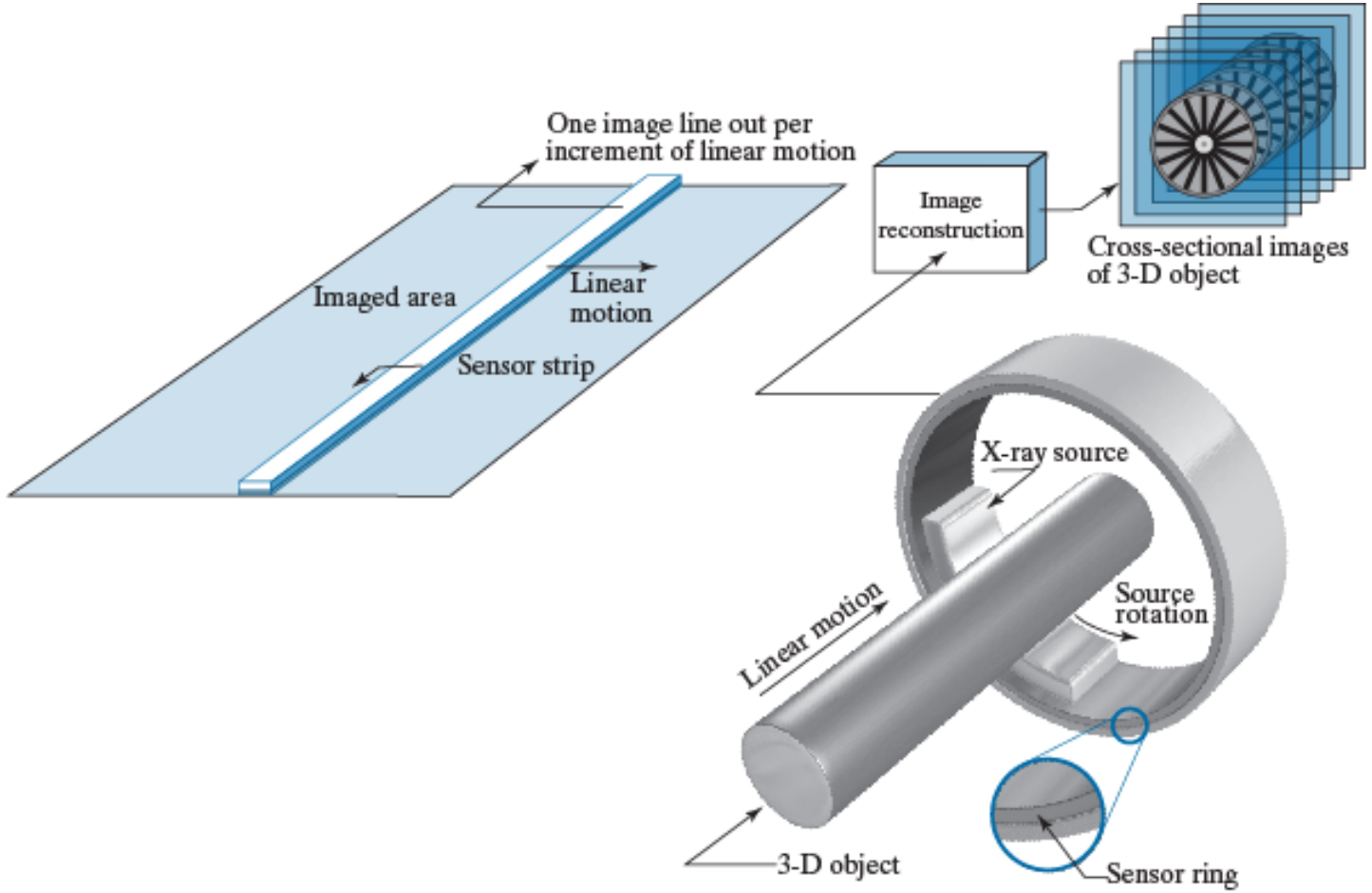



**FIGURE 2.13** Combining a single sensing element with mechanical motion to generate a 2-D image.

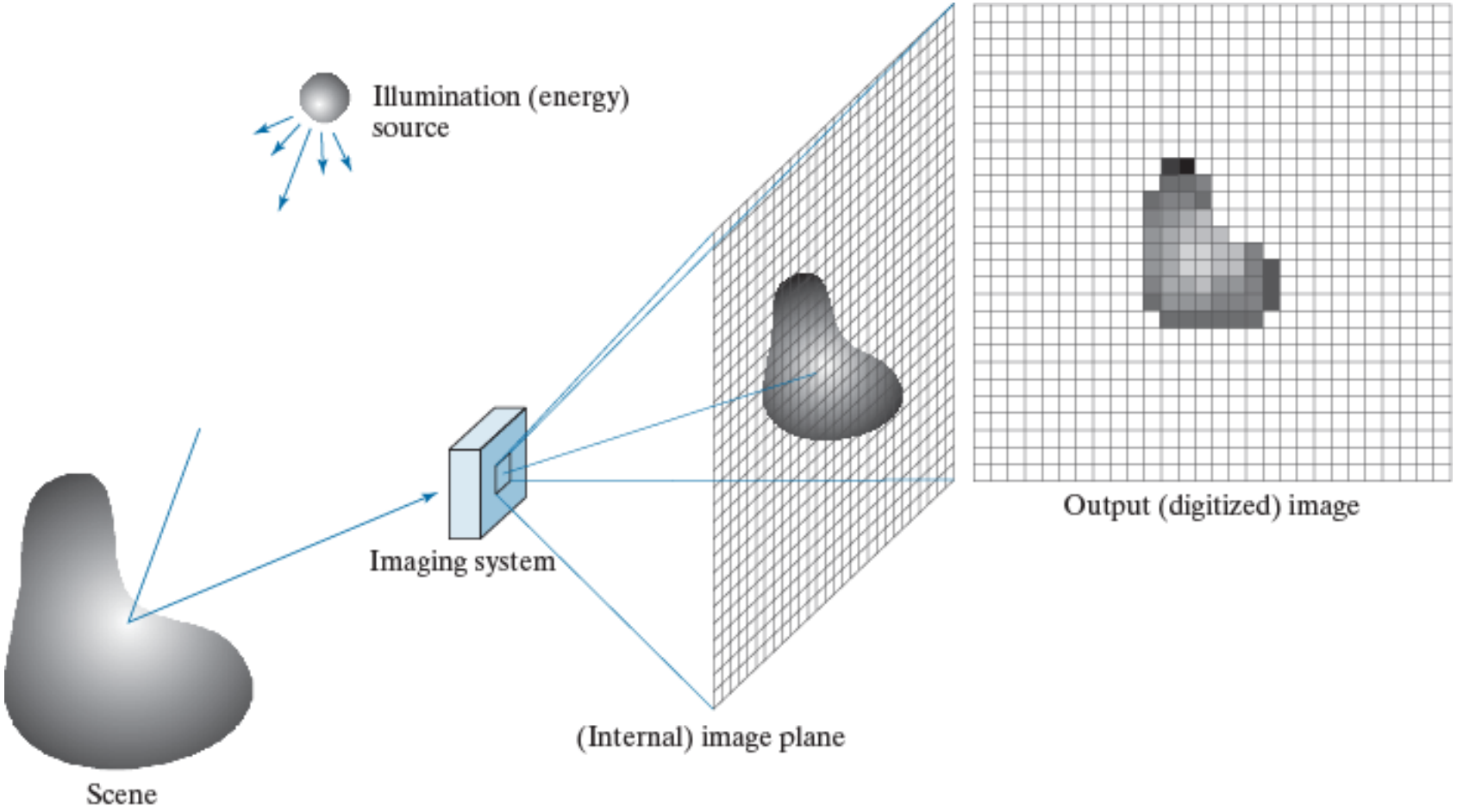
# Image Acquisition using Line Sensor




a b

**FIGURE 2.14** (a) Image acquisition using a linear sensor strip. (b) Image acquisition using a circular sensor strip.

# Image Acquisition using Array Sensor




a b c d e

**FIGURE 2.15** An example of digital image acquisition. (a) Illumination (energy) source. (b) A scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

# Image Model

***f*(*x*,*y*)** can be characterized by two components:

1. The amount of source light incident on the scene being viewed
   Illumination component = ***i*(*x*,*y*)**
2. Amount of light reflected by the objects in the scene
   Reflectance component = ***r*(*x*,*y*)**

$$f(x, y) = i(x, y)r(x, y)$$

$$0 \leq i(x, y) < \infty$$

$$0 \leq r(x, y) \leq 1$$

# Image Model

Reflectance is bounded by 0 (total absorption) and 1 (total reflectance)
The nature of ***i*(*x*,*y*)** is determined by the light source and ***r*(*x*,*y*)** is determined by the characteristics of the objects in a scene.

***i*(*x*,*y*):**
Clear Day – 90,000 lm/m$^2$
Cloudy day – 10,000 lm/m$^2$
Full moon – 0.01 lm/m$^2$
Commercial office - 1000 lm/m$^2$

Typical values of ***r*(*x*,*y*):**
Black Velvet - 0.01
Stainless Steel - 0.65
Flat-White wall paint - 0.80
Silver plated metal - 0.90
Snow - 0.93

# Gray Scale

Intensity of monochrome image *f* at coordinates (*x*,*y*) $l = f(x_0, y_0)$

$$L_{\min} \le l \le L_{\max}$$

$$L_{\min} = i_{\min} r_{\min}$$

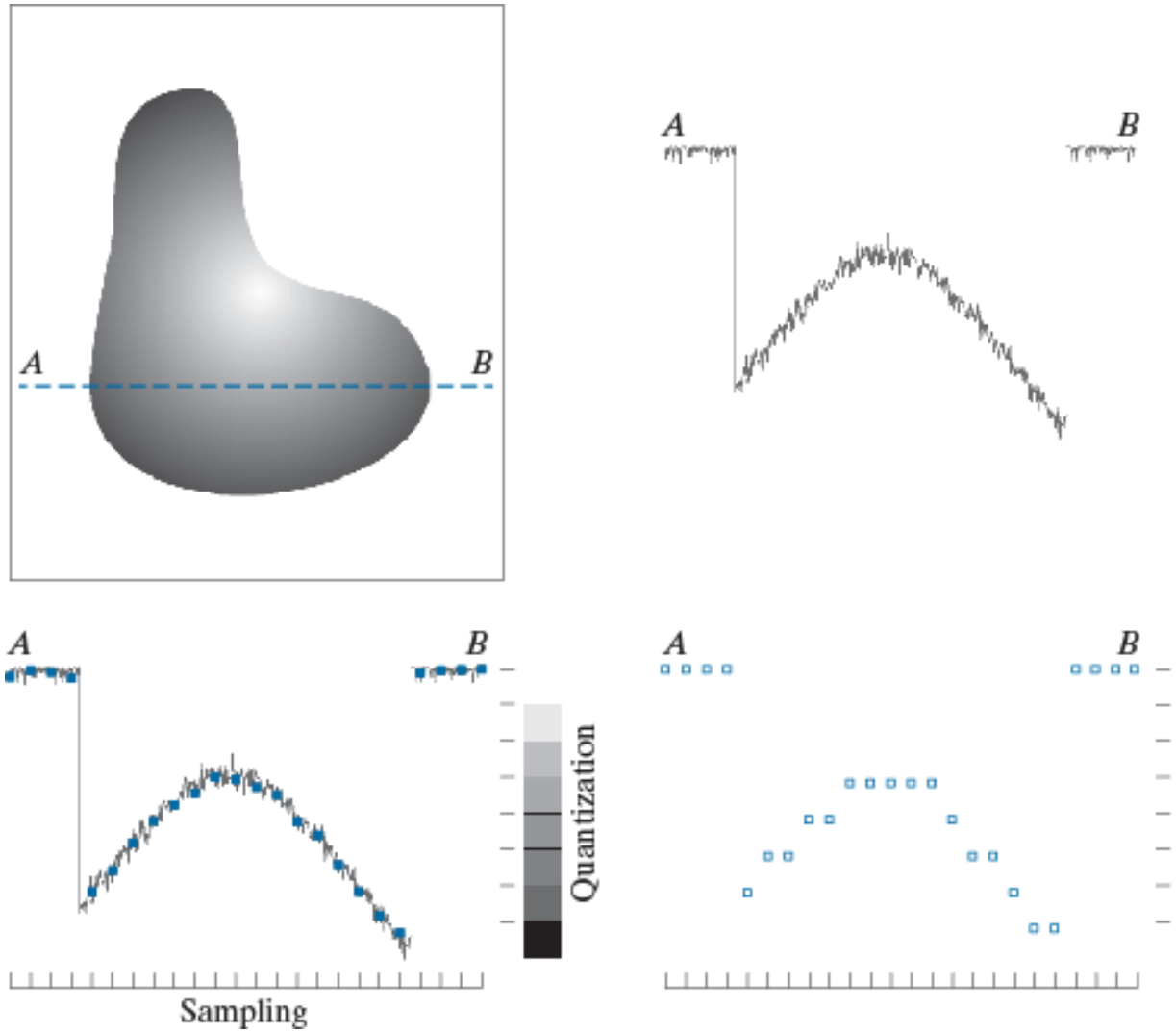
$$L_{\max} = i_{\max} r_{\max}$$

The interval $[L_{\min} \quad L_{\max}]$ is called the gray scale

Commonly, [0, L-1] interval is used.

*0 > Black, & L-1 > White*

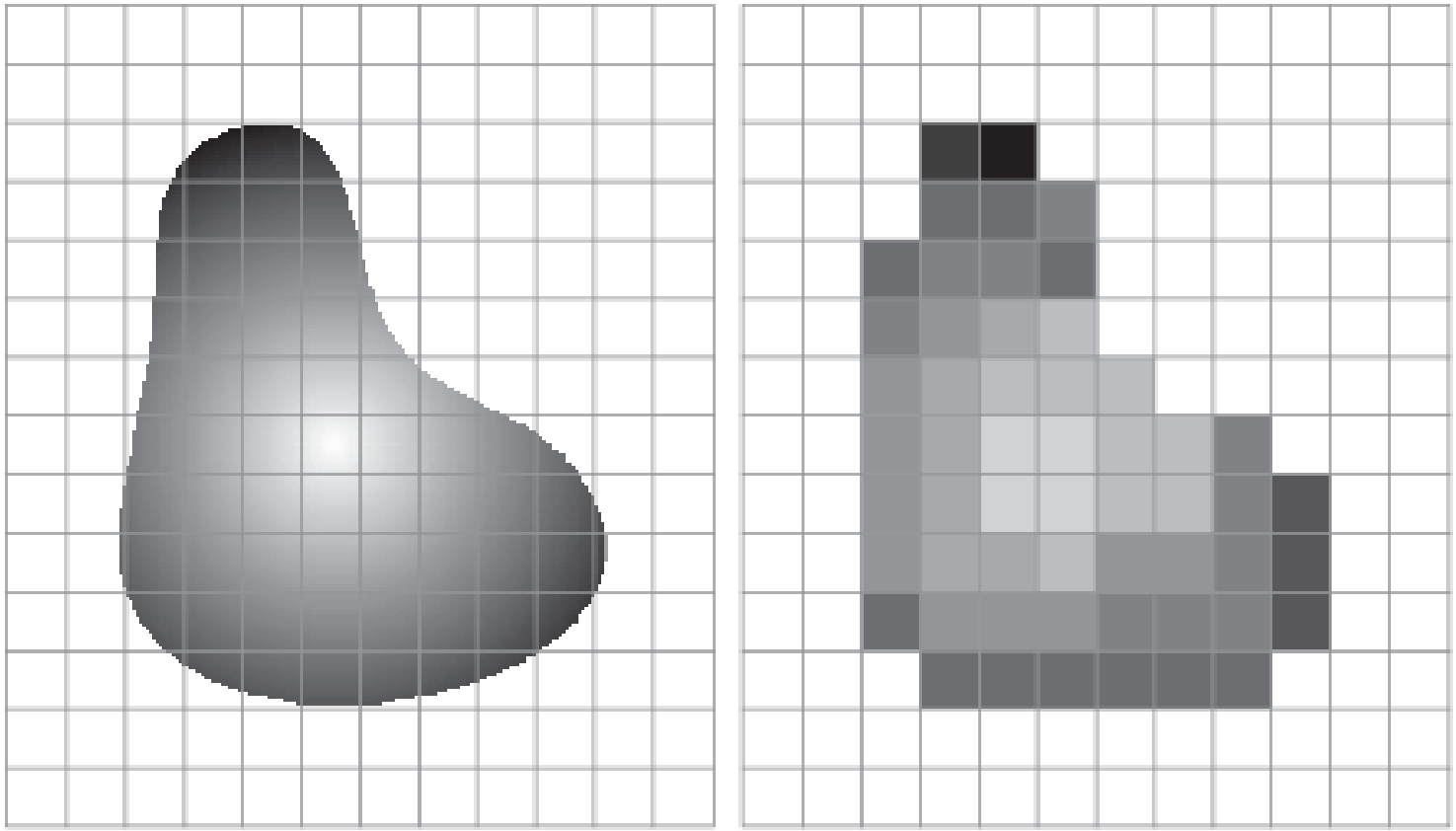
# Image Sampling and Quantization




a b
c d

**FIGURE 2.16** (a) Continuous image. (b) A scan line showing intensity variations along line *AB* in the continuous image. (c) Sampling and quantization. (d) Digital scan line. (The black border in (a) is included for clarity. It is not part of the image).

# Image Sampling and Quantization



a b

**FIGURE 2.17** (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.